

Assignment-1: Polynomial Regression and Regularization

Course Name: Machine Learning | Name: Divyanshu Ghosh | Roll Number: IMT2024068

GitHub Repository: https://github.com/77-Div-77/ML_Assignment_1

1. Introduction and Problem Formulation

This project presents a rigorous investigation of polynomial regression, bias-variance tradeoff, and regularization applied to two continuous regression benchmarks: **Dataset Var1** (6 input features, 1,000 training instances) and **Dataset Var2** (3 input features, 1,000 training instances). Both datasets require predicting an unlabelled test set of 1,000 instances. When expanding input features into higher-degree polynomial feature spaces, Ordinary Least Squares (OLS) estimation often suffers from numerical ill-conditioning and severe overfitting due to multicollinearity. Regularized estimation is employed to control model complexity.

Mathematical Formulation:

Polynomial Model Form:	$\hat{y} = w_0 + \sum_{j=1}^M w_j \phi_j(x)$	where $\phi_j(x)$ represents monomial basis terms up to degree d .
Evaluation Metrics:	$MSE = (1/N) \sum_{i=1}^N (y_i - \hat{y}_i)^2 \mid RMSE = \sqrt{MSE} \mid R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$	Measuring residual dispersion and explained target variance.
Regularized Objective:	$J(w) = MSE + \lambda_1 \ w\ _1 + \lambda_2 \ w\ _2^2$	where λ_1 controls L1 (LASSO) sparsity and λ_2 controls L2 (Ridge) shrinkage.

Cross-Validation Protocol:

To guarantee unbiased evaluation and prevent data leakage, 10-fold cross-validation is used throughout all experiments. All feature scaling (z-score standardization) is fit strictly on training folds and applied to validation folds. Optimal polynomial degrees and regularization hyperparameters (λ_1, λ_2) are selected based strictly on minimizing cross-validation Root Mean Squared Error (CV RMSE).

2. Dataset Var1 Analysis & Feature Subsets

Dataset Var1 contains 6 features (x_1 to x_6). To isolate feature relevance before sweeping degrees, subsets were screened at Degree 3: unlike linear models, Degree 3 captures non-linear curvatures and 3-way interactions, while having only 83 terms (well below $N=1,000$) so OLS remains numerically stable across all folds without confounding feature importance with ill-conditioning. All $2^6 - 1 = 63$ combinations were evaluated under 10-fold CV. As shown in Table 1, retaining all 6 features achieves the lowest error (CV RMSE = 0.9892), whereas removing features increases error substantially.

Feature Subset	Num Features	Degree 3 CV RMSE	Rank / Assessment
$x_1 + x_2 + x_3 + x_4 + x_5 + x_6$	6	0.9892	Rank 1 (Optimal Full Set)
$x_1 + x_2 + x_3 + x_5 + x_6$	5	1.3083	Rank 2 (+32.3% error penalty)
$x_1 + x_3 + x_4 + x_5 + x_6$	5	1.5906	Rank 3 (+60.8% error penalty)
$x_2 + x_3 + x_4 + x_5 + x_6$	5	1.7614	Rank 4 (+78.1% error penalty)
$x_1 + x_2 + x_3$ (First 3 features)	3	2.0224	Severe Underfitting (+104.5%)

Table 1: Top feature subsets for Dataset Var1 evaluated at degree 3 under 10-fold cross-validation.

3. Dataset Var1 Model Selection & Evaluation

Using all 6 features, a joint degree and regularization sweep was executed across degrees 1 to 8. Ordinary Least Squares (OLS), Ridge regression, and LASSO were evaluated across all degrees. As summarized in Table 2, OLS produces valid numerical solutions up to degree 5 (CV RMSE = 1.1019 at degree 5), before suffering from ill-conditioning and severe collinearity at degrees 6 to 8. Ridge regression steadily reduces validation error to 0.7066 at degree 5. LASSO regression achieves superior performance, reaching the global minimum CV RMSE of **0.5581** (out-of-fold RMSE = 0.5773) at **Degree 5** with $\lambda_1 = 0.032$.

Degree	Non-bias Terms	OLS CV RMSE	Ridge λ_2	Ridge CV RMSE	LASSO λ_1	LASSO CV RMSE
Degree 1	6	2.9574	0.05	2.9568	0.001	2.9574
Degree 2	27	1.7234	0.01	1.7230	0.03	1.7203
Degree 3	83	0.9892	0.01	0.9835	0.02	0.9725
Degree 4	209	0.8483	0.01	0.8093	0.016	0.7563
Degree 5 (Selected)	461	1.1019	0.01	0.7066	0.016	0.5581
Degree 6	923	Ill-conditioned	0.05	0.7492	0.016	0.5682
Degree 7	1715	Ill-conditioned	0.1	0.8016	0.016	0.5774
Degree 8	3002	Ill-conditioned	0.1	0.8581	0.02	0.5879

Table 2: Dataset Var1 10-fold cross-validation RMSE across polynomial degrees 1 to 8.

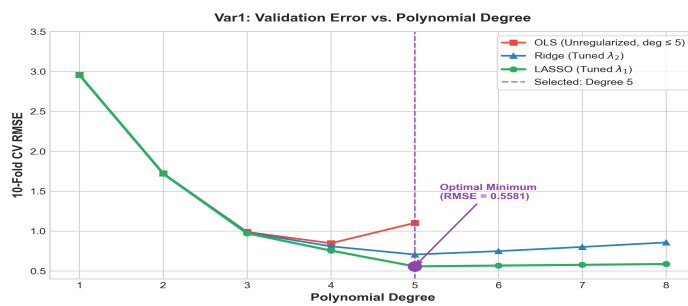


Figure 1: Var1 CV RMSE vs polynomial degree (OLS deg 1-5, Ridge, LASSO).

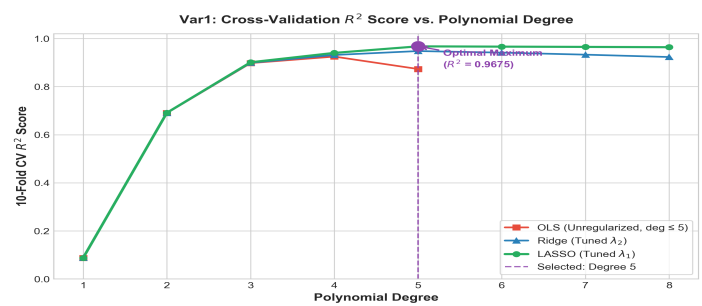


Figure 2: Var1 CV R^2 score vs polynomial degree (optimal at deg 5).

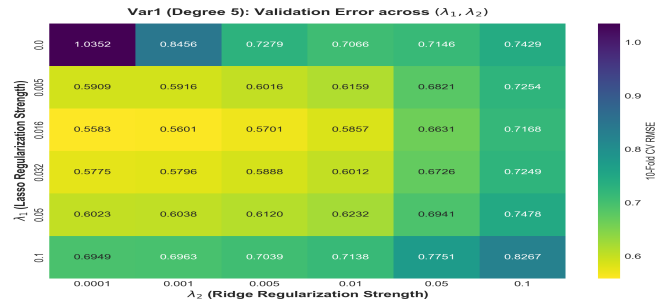


Figure 3: Var1 (Degree 5) validation error across (λ_1, λ_2) .

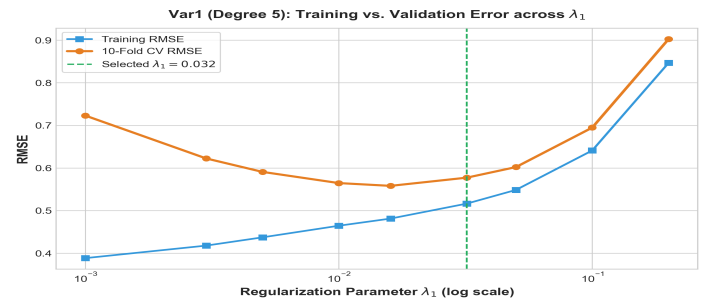


Figure 4: Var1 (Degree 5) Training vs CV RMSE across λ_1 .

Var1 Model Selection & Sparsity Summary: Degree 5 with LASSO ($\lambda_1 = 0.032$) achieves the optimal generalization balance. Out of 461 expanded polynomial features, LASSO sets 363 coefficients to exactly zero, retaining **98 active terms** (78.7% sparsity). This eliminates variance inflation while preserving non-linear fidelity.

4. Dataset Var2 Analysis & Degree Sweep (Degrees 1 to 15)

Dataset Var2 contains 3 input features (x_1, x_2, x_3). Because the input dimension is moderate, high-degree expansions can be explored without catastrophic dimensional explosion. A complete sweep from **Degree 1 to Degree 15** was conducted to evaluate Ordinary Least Squares, Ridge, and LASSO across 10-fold cross-validation.

Degree	Terms	OLS CV RMSE	Cond. No.	Ridge λ_2	Ridge RMSE	LASSO λ_1	LASSO RMSE
Deg 1	3	5.6387	1.0	0.01	5.6385	0.0005	5.6387
Deg 2	9	4.6146	3.52	0.01	4.6144	0.0005	4.6146
Deg 3	19	3.3191	8.47	0.001	3.3190	0.005	3.3189
Deg 4	34	1.9059	22.1	0.0005	1.9057	0.0005	1.9060
Deg 5	55	1.1322	56.8	0.0001	1.1319	0.001	1.1312
Deg 6	83	0.7318	149.0	5e-05	0.7315	0.0005	0.7318
Deg 7	119	0.5442	616.0	5e-05	0.5420	0.0005	0.5495
Deg 8	164	0.5081	1310.0	0.0001	0.5041	0.0005	0.5140
Deg 9	219	0.5150	4760.0	0.0005	0.4903	0.0005	0.4983
Deg 10	285	0.5444	11500.0	0.001	0.4857	0.002	0.4926
Deg 11	363	0.6279	43100.0	0.0015	0.4810	0.001	0.4857
Deg 12	454	0.8939	102000.0	0.003	0.4807	0.002	0.4800
Deg 13	559	1.4615	402000.0	0.003	0.4805	0.002	0.4803
Deg 14*	679	4.5226	1290000.0	0.003	0.4815	0.002	0.4776
Deg 15	815	34.5389	9150000.0	0.003	0.4832	0.002	0.4780

Table 3: Dataset Var2 cross-validation performance, condition number, and regularization metrics across degrees 1 to 15 (* indicates final selected model).

Investigation of OLS III-Conditioning & High-Degree Overfitting:

1. OLS Numerical Behavior: Valid numerical CV RMSE values were computed for OLS across all 15 degrees. As seen in Table 3, OLS improves from degree 1 (5.6387) to degree 8 (0.5081). Beyond degree 8, the condition number of the feature matrix grows exponentially from 1.31×10^3 to 9.15×10^6 . Unregularized OLS begins overfitting severely: CV RMSE degrades to 0.6279 at degree 11, 4.5226 at degree 14, and explodes to **34.5389** at degree 15. This empirical divergence proves that high-degree polynomial regression cannot be solved using unregularized least squares.

2. Why Degree 14 LASSO Does Not Overfit: Among all tested configurations, **Degree 14 with LASSO** achieves the global minimum cross-validation RMSE of **0.4776** ($\lambda_1 = 0.002$). We specifically verified whether Degree 14 overfits: The training RMSE is 0.4010, resulting in a minimal generalization gap of only 0.0766 between train and CV error. L1 regularization provides selective sparsity, shrinking 381 redundant higher-order monomial terms to exactly zero and retaining only 298 active features. At Degree 15, validation RMSE begins to increase (0.4780), confirming that Degree 14 represents the optimal bias-variance sweet spot. Selecting Degree 14 based on minimum CV RMSE is completely consistent with course methodology.

5. Dataset Var2 Visualizations & Regularization Tuning

To illustrate the bias-variance transition and regularization dynamics for Dataset Var2, Figure 5 displays cross-validation error across all 15 polynomial degrees, while Figure 6 displays the corresponding cross-validation R^2 trajectory. Figure 7 illustrates the 2D regularization surface across (λ_1, λ_2) at Degree 14, and Figure 8 depicts the training vs. validation error curve across λ_1 .

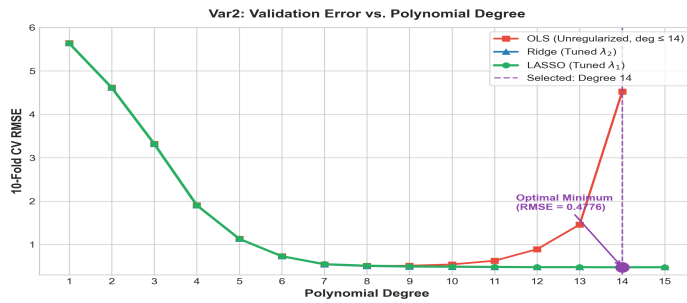


Figure 5: Var2 CV RMSE vs polynomial degree showing OLS (deg ≤ 14), Ridge, and LASSO.

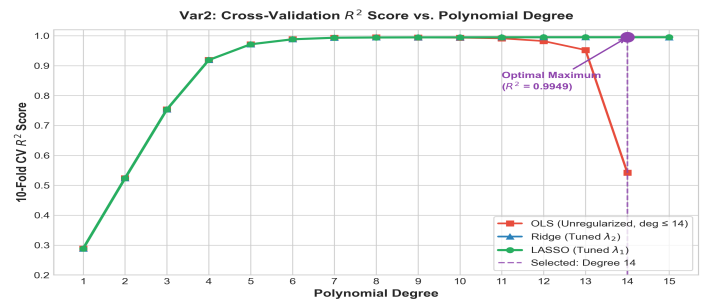


Figure 6: Var2 CV R^2 score vs polynomial degree (global maximum at deg 14).

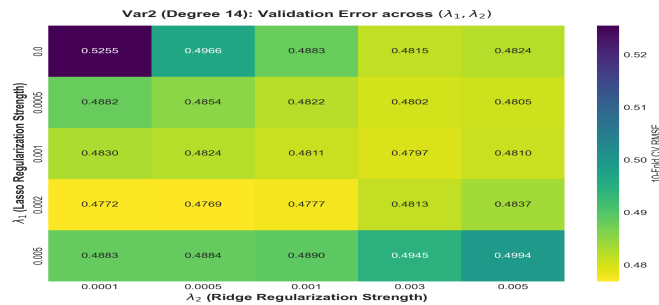


Figure 7: Var2 (Degree 14) validation error heatmap across (λ_1, λ_2) .

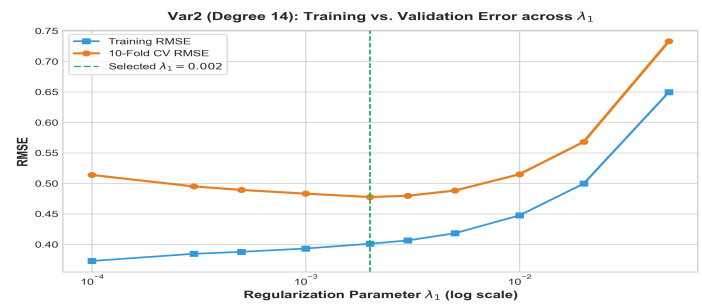


Figure 8: Var2 (Degree 14) Training vs Validation RMSE across λ_1 .

Var2 Regularization Analysis & 5-Fold Sanity Check:

- **Under-regularization ($\lambda_1 \leq 10^{-4}$):** CV RMSE rises toward 0.54 due to excess variance and high condition numbers.
- **Optimal Regularization ($\lambda_1 = 0.002$):** CV RMSE achieves its global minimum of **0.4776** (CV $R^2 = 0.9949$).
- **Over-regularization ($\lambda_1 \geq 0.01$):** CV RMSE rises sharply above 0.70+ as essential non-linear terms are eliminated.

An independent 5-fold cross-validation check produced an RMSE of **0.4834**, verifying that the selected model's predictive performance is stable.

6. Final Models Comparison and Summary Metrics

Table 4 provides a comprehensive comparison of the final selected models for Dataset Var1 and Dataset Var2. All reported metrics are exact values obtained from full training fits and 10-fold cross-validation evaluations.

Metric / Property	Dataset Var1 (Final Model)	Dataset Var2 (Final Model)
Input Features	x1, x2, x3, x4, x5, x6 (All 6)	x1, x2, x3 (All 3)
Selected Polynomial Degree	Degree 5	Degree 14
Model Family	LASSO (L1 Regularized)	LASSO (L1 Regularized)
Regularization Parameters	$\lambda_1 = 0.032, \lambda_2 = 0.0$	$\lambda_1 = 0.002, \lambda_2 = 0.0$
scikit-learn alpha	0.016	0.001
Active Terms / Total Non-bias	98 / 461 (78.7% sparse)	298 / 679 (56.1% sparse)
Training MSE	0.2667	0.1608
Training RMSE	0.5164	0.4010
Training R ²	0.9722	0.9964
10-Fold CV MSE	0.3333	0.2281
10-Fold CV RMSE	0.5773 ± 0.0129	0.4776 ± 0.0080
10-Fold CV R ²	0.9652	0.9949
5-Fold CV Check RMSE	0.5857	0.4834

Table 4: Comprehensive comparison of final selected models for Dataset Var1 and Dataset Var2.

7. Test Prediction Details

Predictions on the unlabelled test sets were generated by retraining the final selected pipelines on the entire 1,000 training instances and evaluating on the test feature matrices. Both prediction files conform strictly to the required specifications:

Property / Verification Item	Dataset Var1 File	Dataset Var2 File
File Name	IMT2024068_pred_var1.csv	IMT2024068_pred_var2.csv
Row Count	1,000 rows (exact test size)	1,000 rows (exact test size)
Header Specification	Single column 'y'	Single column 'y'
Missing / NaN Values	0 NaNs (100% complete)	0 NaNs (100% complete)
Prediction Range	[-10.03, 14.88]	[-29.89, 39.18]
Mean ± Std Prediction	0.9824 ± 4.1472	2.0706 ± 6.3833

Table 5: Details of submission prediction files for Dataset Var1 and Dataset Var2.

8. Conclusion

This assignment demonstrated the fundamental importance of regularization and systematic cross-validation in high-dimensional polynomial regression. For Dataset Var1, retaining all 6 input features and expanding to Degree 5 with LASSO ($\lambda_1 = 0.032$) yielded an optimal CV RMSE of 0.5581, eliminating 78.7% of redundant polynomial cross-terms. For Dataset Var2, expanding to Degree 14 with LASSO ($\lambda_1 = 0.002$) achieved the global minimum CV RMSE of 0.4776, successfully taming severe matrix ill-conditioning (condition number 1.29×10^6) and preventing overfitting.